# Session 7 Homework: Scoreboard and Strategy 3

Computer Vision and Speech Recognition — Slide 30

Author: Kacper Dudzik

Repository: https://github.com/KacperDudzik/CV_session

Code: session7/homework.py (run with python homework.py)

Supporting code: session7/session7.py

Raw results: _output/homework_results.json

## 1. Scoreboard (Sessions 5–7)

Metric: lesion-level macro-F1. Session 7 models were trained in the homework run; test macro-F1 is scored once after selecting the best validation checkpoint.

| Model | Val macro-F1 | Test macro-F1 |
|---|---|---|
| Session 5 - random forest (33 features, course) | 0.534 | 0.495 |
| Session 5+ midterm SVM (all_features, oversample) | 0.567 | 0.597 |
| Session 6 - MLP 33-64-3 (33 features) | 0.493 | 0.472 |
| Session 7 - SimpleCNN (scratch) | 0.522 | 0.510 |
| Session 7 - ResNet-18 frozen + linear head (strategy 1) | 0.629 | 0.560 |
| Session 7 - ResNet-18 fine-tune layer4+fc (strategy 2) | 0.645 | 0.613 |
| Session 7 - ResNet-18 fine-tune layer3+layer4+fc | 0.660 | 0.581 |
| Session 7 - ResNet-18 full fine-tune (strategy 3) | 0.644 | 0.564 |

Best model: Session 7 - ResNet-18 fine-tune layer4+fc (strategy 2) — val 0.645, test macro-F1 = 0.613.

## 2. DenseNet vs ResNet (one sentence)

DenseNet differs from ResNet by concatenating each layer's feature maps to all later layers (dense connectivity) instead of adding residual skip connections, which encourages feature reuse and typically needs fewer parameters.

## 3. Optional: full fine-tuning (strategy 3)

Strategy 3 unfreezes every ResNet-18 parameter (~11.2M trainable) and trains at lr = 1e-4 with class-weighted cross-entropy and flips, same protocol as partial fine-tuning.

| Setting | Trainable params | s / epoch | Val F1 | Test F1 |
|---|---|---|---|---|
| Strategy 2 (layer4+fc) | 8,395,267 | 94 | 0.645 | 0.613 |
| Strategy 3 (full network) | 11,178,051 | 161 | 0.644 | 0.564 |

Finding: full fine-tuning did not beat partial layer4 fine-tuning on test; keep strategy 2 as the default.

## 4. Notes

- Session 7 models use lesion-grouped dermoscopic StratifiedGroupKFold (seed=0).

- Midterm SVM uses its own seed=42 lesion split; reported for completeness as best classical.

- Test macro-F1 is computed once per model after selecting the best validation checkpoint.

Device: cpu; seed 0; splits {'train': 3668, 'val': 1048, 'test': 524}.